

AARYAN PRAKASH

I enjoy building technology that solves practical problems while understanding the business and economic context behind it. I'm equally interested in how engineering decisions shape products, businesses, and create real-world impact. I enjoy exploring different areas of computer science and continuously learning through new challenges and projects.



CONTACT

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SKILLS

Programming

Python, Java, C

JavaScript, HTML/CSS, SQL

AI & Machine Learning

Machine Learning, Deep Learning, NLP

Computer Vision, Federated Learning

Technologies

Django, FastAPI, OpenCV, Pandas, MySQL

NumPy, BERT, LightGBM, Git, GitHub

Core Concepts

Data Structures, Algorithms, OOP

Cybersecurity, Network Security

Languages

English, Hindi, Kannada

EXPERIENCE

President, Protocol BMSCE Apr 2025 – May 2026

CSE Department Club, BMSCE

- Lead planning and execution of technical and cultural events across college fests.
- Handled faculty coordination, sponsorships, and team management.

Editor & Journalist, Synergy Weekly May 2025 – May 2026

Weekly Tech Newsletter

- Ran a weekly tech newsletter, overseeing topic selection and writing.
- Guided a small team on editorial direction and tone.

Junior Core, Protocol BMSCE Jun 2024 – Apr 2025

- Organised and marketed technical events across campus.

EDUCATION

Aug 2023 – Present

Bachelor of Engineering, Computer Science

BMS College of Engineering

CGPA: 8.40

2016 – 2023

Middle and High School

DPS Bangalore South

Grade: 90 (XII), 93 (X)

2009 – 2016

National Hill View Public School

PROJECTS

Robust Semi-Supervised Federated Learning-Based Intrusion Detection System Jan 2026 – Present

Federated Learning & Cybersecurity Research Project

- Developing a federated learning-based Intrusion Detection System (IDS) for Software-Defined Networking (SDN)-enabled IoT environments.
- Designing a semi-supervised architecture combining Autoencoder-based anomaly detection, trust-based malicious client filtering, and robust aggregation to improve resilience against poisoning attacks.
- Optimizing the system using cosine similarity evaluation and lightweight communication techniques to support non-IID IoT data while reducing computational and communication overhead.
- Conducting literature review on Federated Learning, adversarial machine learning, semi-supervised learning, and IoT security to identify research gaps and guide system design.

Fake News Detection using BERT & LightGBM Sep 2025 – Dec 2025

Machine Learning & NLP Project

- Built a fake news classification system using BERT to generate contextual embeddings from news articles.
- Fine-tuned BERT to extract semantic representations for improved text classification performance.
- Trained a LightGBM classifier on BERT embeddings to achieve efficient and accurate prediction.
- Evaluated model performance using accuracy, precision, recall, and F1-score.
- Designed a modular pipeline separating feature extraction from classification for improved scalability and maintainability.

Drowsiness Detection System

Computer Vision Project

Mar 2026 – May 2026

- Developed a real-time drowsiness detection system using OpenCV, dlib, and SciPy.
- Implemented facial landmark detection using dlib's 68-point predictor to accurately localize eye regions in webcam video.
- Computed Eye Aspect Ratio (EAR) from facial landmarks to detect prolonged eye closure and signs of fatigue.
- Triggered visual alerts when EAR remained below a threshold across consecutive frames, reducing false positives from normal blinking.
- Built a lightweight real-time pipeline for video capture, face detection, feature extraction, and drowsiness classification.

AI-Powered Flashcard Generator

Full-Stack Web Application

Oct 2024 – Jan 2025; Updated 2026

- Built a full-stack flashcard application using React, Tailwind CSS, Supabase, and PostgreSQL with email authentication, protected routing, and row-level security.
- Extended the application with an AI flashcard-generation pipeline using PDF.js and Tesseract.js to extract text from notes, PDFs, and images, with Groq's Llama 3.3 generating term-definition pairs through Supabase Edge Functions.
- Implemented multiple-choice quizzes with per-user attempt history, scoring, and flashcard-set management.
- Added public read-only sharing, image uploads, print-friendly layouts, and deployed the application using Vercel and Supabase with version-controlled SQL migrations.

Formula 1 Analytics Dashboard

Full-Stack Data Visualization Project

Sep 2025 – Dec 2025; Updated 2026

- Built a full-stack F1 race analytics dashboard using FastAPI, JavaScript, and Plotly.js, with 8 REST endpoints for interactive data exploration.
- Engineered a data pipeline using the FastF1 API for lap times, pit stops, tyre strategies, gaps, sectors, telemetry, and weather across F1 seasons.
- Designed interactive dashboard views with driver comparisons, team-livery-accurate visualizations, track maps, and telemetry-based speed, throttle, and brake analysis.
- Deployed the application on Render with in-memory and on-disk caching to improve performance when loading large telemetry datasets.

CERTIFICATIONS

Introduction to Responsible AI

Google Cloud Skills Boost

Supply Chain Digitization

NPTEL

Research Methodology

NPTEL

Artificial Intelligence Foundation

Infosys Springboard

PROFESSIONAL SKILLS

- Leadership

Team Management

Technical Writing

Public Speaking

Sponsorship